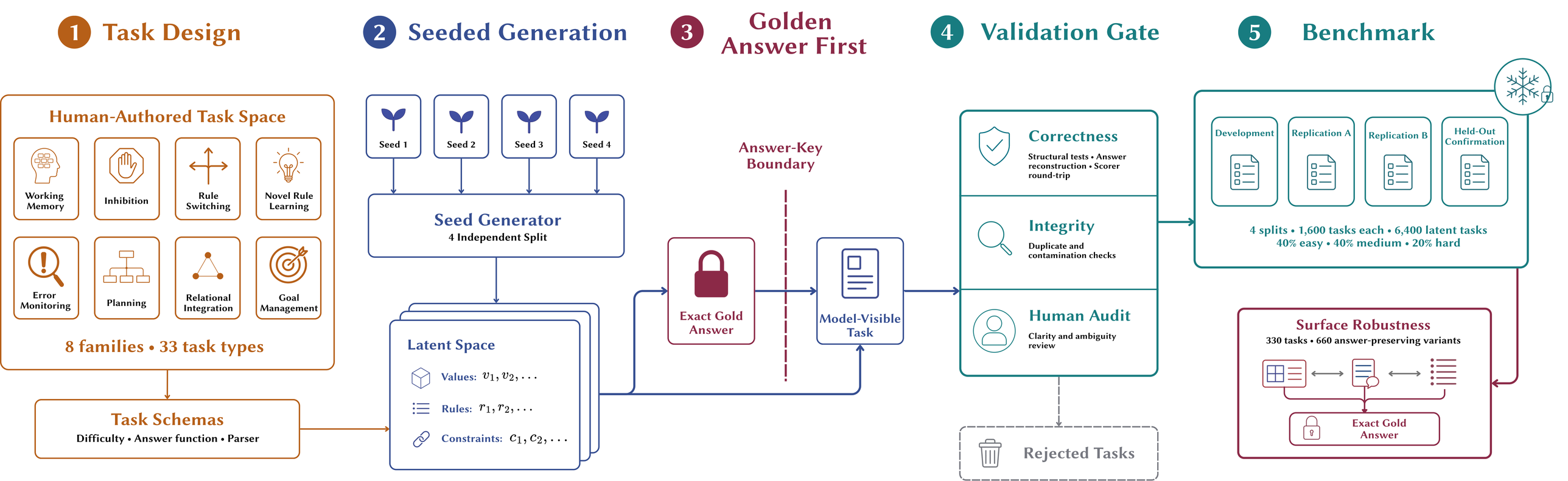

1 Task Design
2 Seeded Generation
3 Golden Answer First
4 Validation Gate
5 Benchmark
Human-Authored Task Space
Working Memory
Inhibition
Rule Switching
Novel Rule Learning
Error Monitoring
Planning
Relational Integration
Goal Management
8 families • 33 task types
Task Schemas
Difficulty • Answer function • Parser
Seed 1
Seed 2
Seed 3
Seed 4
Seed Generator
4 Independent Split
Latent Space
Values: $v_1, v_2, \ldots$
Rules: $r_1, r_2, \ldots$
Constraints: $c_1, c_2, \ldots$
Answer-Key Boundary
Exact Gold Answer
Model-Visible Task
Correctness
Structural tests • Answer reconstruction • Scorer round-trip
Integrity
Duplicate and contamination checks
Human Audit
Clarity and ambiguity review
Rejected Tasks
Development
Replication A
Replication B
Held-Out Confirmation
4 splits • 1,600 tasks each • 6,400 latent tasks
40% easy • 40% medium • 20% hard
Surface Robustness
330 tasks • 660 answer-preserving variants
Exact Gold Answer